Voting Regressor

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Where this fits: Pipeline map steps 8 (Model), 9 (Evaluate). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	**8 Model**	**9 Evaluate**	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Cross-validation; Voting ensembles.
- **Compare with:** Train-test split; Bagging; OOB score; Stacking and blending.

1. Overview

Key point: A voting regressor trains several regressors on the same data and predicts the mean of their outputs, or a weighted mean if we give weights.

The **voting regressor** (G-2097) is the regression version of the voting classifier (see the voting classifier). Only the aggregation step changes: instead of a vote, we take the **mean** (G-1203) of the **base models'** (G-260) outputs. This Note covers:

- the core idea (section 2) and a demo with curves (section 3);
- scikit-learn's `VotingRegressor` on the Boston housing data, with `weights` (sections 4.1 to 4.3);
- voting over one algorithm with different settings (section 4.4);
- why the cross-validation folds must be shuffled on this data (section 4.5).

The Notebook (`ML-098-voting-regressor.ipynb`) runs every experiment. The Dash app `app.py` lets us tick base regressors and redraws their curves next to the voting regressor's.

2. The core idea

Key point: Train every regressor on the same data; to predict, average their numerical outputs.

We have base models M1, M2 and M3, all regression algorithms, trained on the same dataset D. For a new query point x_q, each returns a number, and the voting regressor returns their mean (regression: the mean works an example). With weights, it returns the weighted mean, exactly as in soft voting (see weights in the voting classifier).

1. **In words:** add the base models' predictions and divide by how many there are.
2. **Formula:** with n base models predicting $f_1, \ldots, f_n$ for the query point, and optional weights $w_1, \ldots, w_n$ ($\sum$ means "add up the terms for $i = 1, \ldots, n$"),

$$\hat{y} = \frac{1}{n} \sum_{i=1}^{n} f_i$$

   Or, with weights:

$$\hat{y} = \frac{\sum_{i=1}^{n} w_i f_i}{\sum_{i=1}^{n} w_i}$$

3. **Example:** three models predict 0.5, 0.8 and 0.55. The vote is:

$$\frac{0.5 + 0.8 + 0.55}{3} = 0.617$$

With weights (1, 2, 1) it is:

$$\frac{1 \times 0.5 + 2 \times 0.8 + 1 \times 0.55}{1 + 2 + 1}$$

$$= \frac{0.5 + 1.6 + 0.55}{4} = 0.663$$

This is pulled towards the second model.

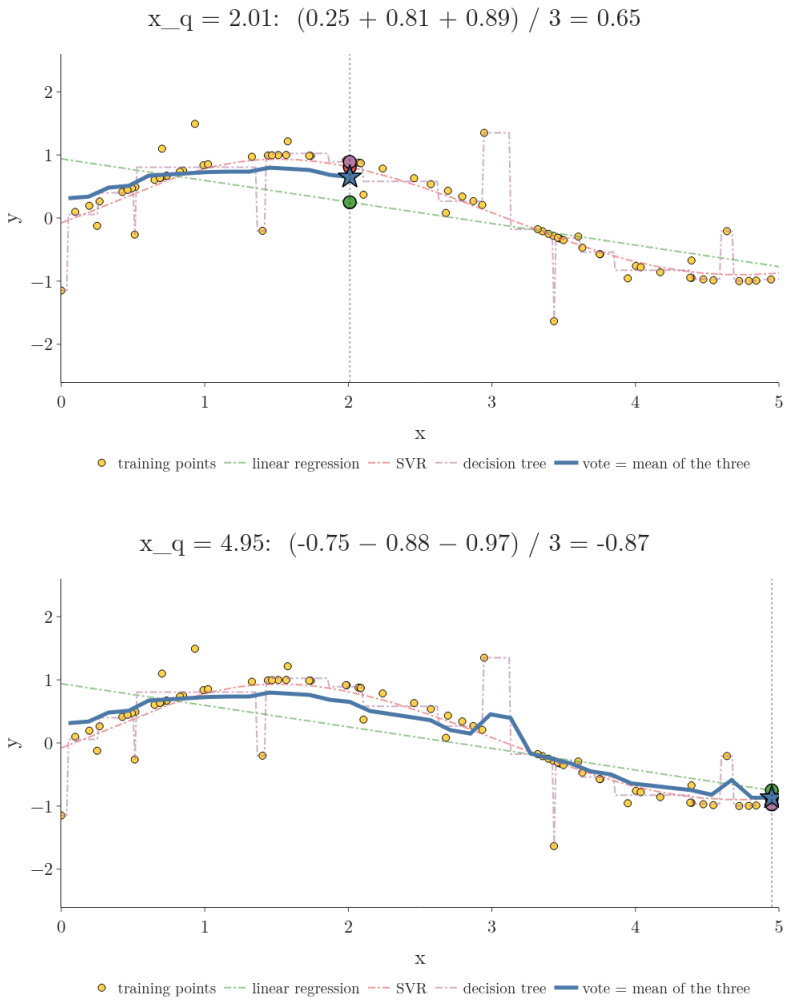


Figure 1: A query point sweeps across the sine data of section 3; at each position the three base regressors give a number each (coloured dots), and the vote is their mean (blue star)

In Figure 1, watch the star: at every x_q it sits at the mean of the three dots, and its trail is the voting regressor's curve.

3. Seeing it on a curve

> **Key point:** The voting regressor's curve runs between its base models' curves and smooths out the extremes of each. Its score is never below the average score of its members.

The demo data is a sine wave of 80 points between 0 and 5, with every fifth point pushed up or down at random: a non-linear pattern with a few outliers. We train on 72 points and keep 8 for testing. Figure 2 adds three base regressors one at a time and redraws the voting regressor after each.

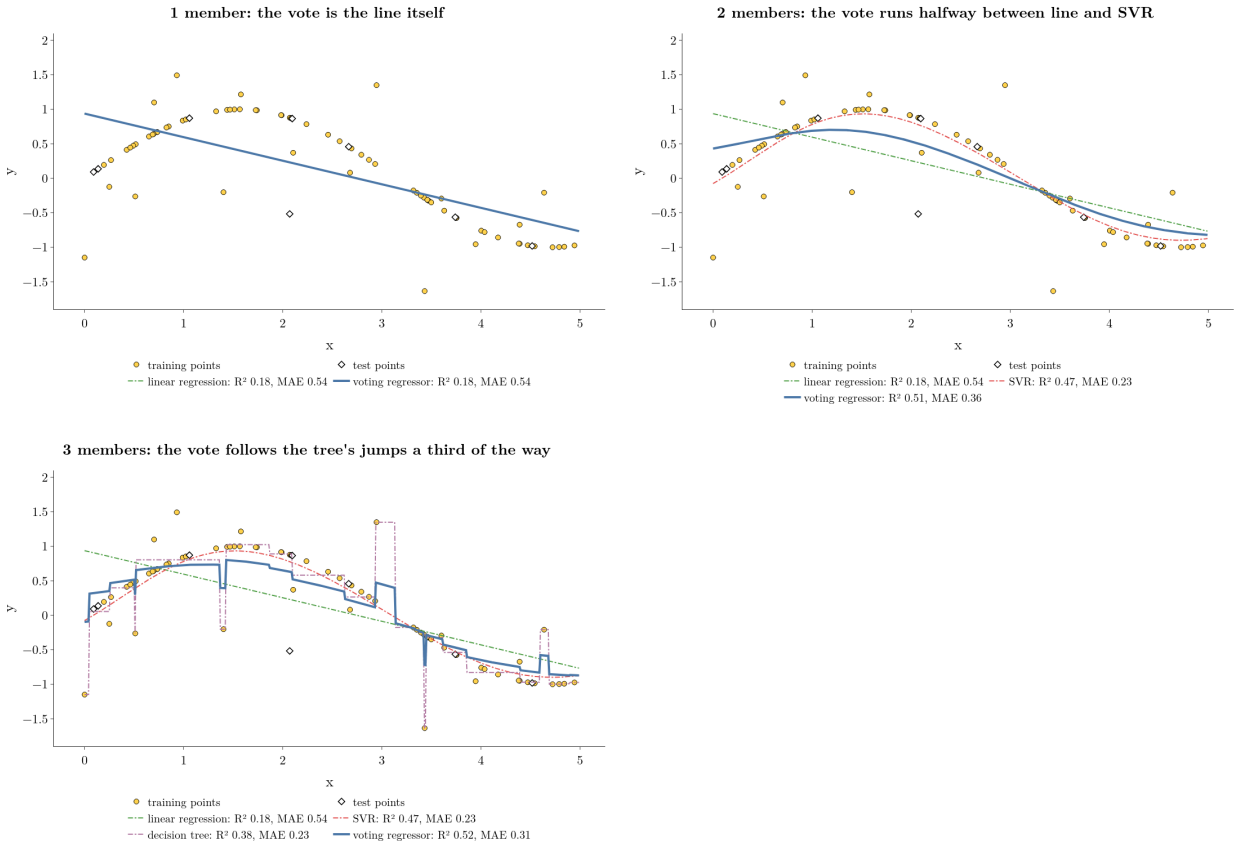


Figure 2: Base regressors (dash-dot) added one at a time on the noisy sine data, and the voting regressor (solid blue) after each addition: the line alone, then halfway between line and SVR, then the mean of line, SVR and tree. Legends give R^2 and MAE on the 8 test points

Think of three friends guessing a distance: one always guesses short, one guesses well, one guesses wildly. The average of their guesses is rarely the single best guess, but it is never as bad as the worst one, and it is steadier than the wild guesser.

The picture shows the same thing:

- **Linear regression** fits a straight line, which cannot follow a wave.
- **SVR** (support vector regression, a model that fits a smooth curve through the points, taught in SVM intuition) follows the wave well.
- **A decision tree** (depth 5) jumps to reach many single points, outliers included.
- **The voting curve** runs between them. With two models it sits exactly halfway between the line and the SVR curve; with three, it follows the tree's jumps only a third of the way, so it is much smoother than the tree.

A score from 8 test points depends heavily on which 8 points we drew. So the Notebook draws 30 fresh datasets with the same recipe and averages the $\mathbf{R^2}$ **score** (G-1717; 1 means perfect predictions, 0 means no better than always predicting the mean; see R^2) on their test points:

Base models	Each model's mean R^2	Average of the members	Voting regressor (G-2097)
linear regression, SVR	0.45, 0.77	0.61	**0.69**
linear regression, SVR, decision tree	0.45, 0.77, 0.53	0.58	**0.72**

The vote is always well above the average of its members. It does not beat SVR here, because SVR alone already fits this smooth wave almost as well as any model can, and the line drags the average down. Voting pays off most when the members are about equally good and make different mistakes, as on the real data below.

Extra: Why can the vote never be worse than the average member? Take one test point with true value y, and let model i predict f_i. The vote predicts:

$$\bar{f} = \frac{1}{n}\sum_i f_i$$

Its error is the mean of the members' errors:

$$\bar{f} - y = \frac{1}{n}\sum_i (f_i - y)$$

The square of a mean is never larger than the mean of the squares. Try it on two members with $y = 10$, $f_1 = 8$, $f_2 = 13$, one line per quantity:

$$\bar{f} = (8 + 13)/2 = 10.5$$

$$(\bar{f} - y)^2 = 0.5^2 = 0.25$$

Mean squared error of the members:

$$\tfrac{1}{2}\Big[(8 - 10)^2 + (13 - 10)^2\Big]$$

$$= \tfrac{1}{2}(4 + 9) = 6.5$$

Spread around the mean:

$$\tfrac{1}{2}\Big[(8 - 10.5)^2 + (13 - 10.5)^2\Big]$$

$$= \tfrac{1}{2}(6.25 + 6.25) = 6.25$$

The two add up correctly:

$$6.5 - 6.25 = 0.25 \quad \checkmark$$

The same identity for any n members, written with $\sum_{i=1}^{n}$ (add the term for $i = 1, 2, \ldots, n$):

$$(\bar{f} - y)^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(f_i - y)^2 - \frac{1}{n}\sum_{i=1}^{n}(f_i - \bar{f})^2$$

The last term is the spread of the members around their mean, which Krogh and Vedelsby (1995) call the **ambiguity** (G-2155). The more the members disagree, the more the vote gains over the average member. Since R^2 falls as squared error rises, the vote's R^2 is at least the average of the members' R^2 on any test set.

4. VotingRegressor on the Boston housing data

Key point: Three base models with R^2 0.71, 0.73 and 0.67 combine into a voting regressor with R^2 0.81, better than each; weights lift it to 0.82.

4.1 The data and the base models

Key point: 506 districts, 13 features, median house price; linear regression, a decision tree and SVR, each scored by shuffled 10-fold cross-validation.

The Boston housing data (see the Boston data and a first tree for how it is used, and why `load_boston` was removed from scikit-learn) has 506 **observations** (one record each, here a district; one row of the data table) and 13 **features** (input variables, one column each). The **target**, the output we predict, is `MEDV`, the median house price of each district. The Notebook reads it from `data/boston.csv`.

We score each model under three conditions:

- **shuffled folds:** the observations are stored town by town, so we shuffle them before cutting the 10 folds (section 4.5 shows why);
- **five repeats:** the whole 10-fold **cross-validation** (G-510; the data is cut into 10 parts, and the model is tested on each part after training on the other nine) is repeated with 5 different shuffles, and we report the mean;
- **scaled SVR:** SVR measures distances between observations, so it gets standardized features (see the standardization formula) through a pipeline (see what a pipeline is).

Python: Each base model alone.

```
boston = pd.read_csv("data/boston.csv")
X, y = boston.drop(columns="MEDV"), boston["MEDV"]
cv = RepeatedKFold(n_splits=10, n_repeats=5, random_state=0)

estimators = [("lr", LinearRegression()),
              ("dt", DecisionTreeRegressor(random_state=0)),
              ("svr", make_pipeline(StandardScaler(), SVR()))]
```

```
for name, model in estimators:
    scores = cross_val_score(model, X, y,
                             scoring="r2", cv=cv)
    print(name, scores.mean())
```

`scoring="r2"` makes `cross_val_score` report R^2 instead of accuracy. `RepeatedKFold` shuffles the observations before cutting each set of folds.

Model	Linear regression	Decision tree	SVR
Mean R^2	0.71	0.73	0.67

The three models are about equally good, and they work in very different ways: a linear formula, a set of yes/no rules, and a smooth curved surface. Those are the right conditions for a vote: members that are accurate and make different mistakes (see more models, and correlated models).

4.2 The voting regressor

Key point: Passing the same three models to `VotingRegressor` lifts R^2 from the best member's 0.73 to 0.81.

Python: The voting regressor.

```
from sklearn.ensemble import VotingRegressor

vr = VotingRegressor(estimators)
cross_val_score(vr, X, y, scoring="r2", cv=cv).mean()    # 0.81
```

`VotingRegressor` takes the same list of (name, model) tuples as `VotingClassifier`, plus optional `weights` and `n_jobs`. There is no `voting` setting: regression always averages.

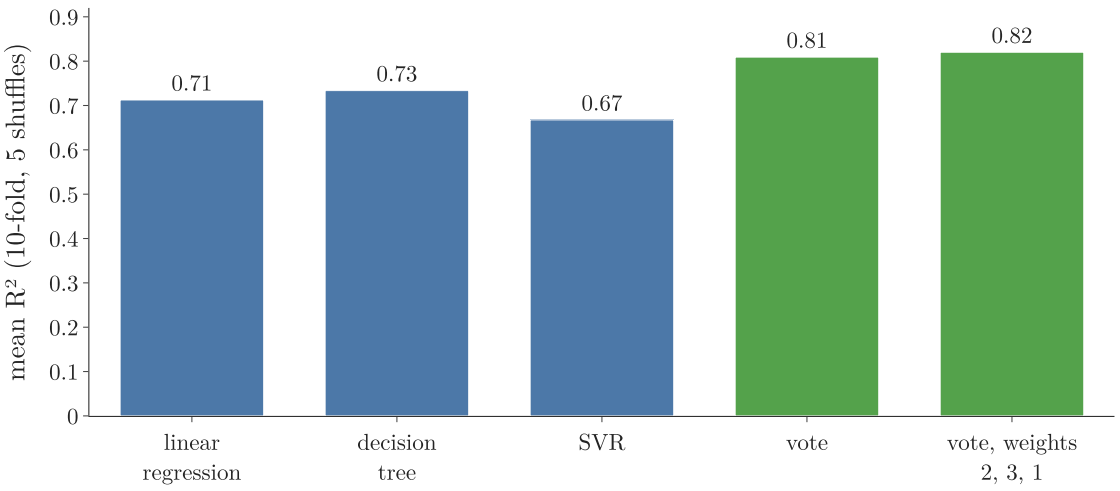


Figure 3: Mean R^2 on the Boston housing data: the three base models, their vote, and the vote with the best weights (2, 3, 1)

In Figure 3, compare the green bars with the blue ones: the plain vote already gains 0.08 over the best member; the weights add only 0.01 more.

The vote scores **0.81**, clearly above every member (0.71, 0.73 and 0.67). Where the tree makes a jump that the line does not, and the SVR smooths both, the errors partly cancel in the average.

`n_jobs=-1` trains the base models in parallel on all CPU cores. On small data like this it does not pay: in the Notebook one fit takes about 0.1 seconds on one core but several seconds with `n_jobs=-1`, because starting parallel work has a cost of its own (scikit-learn User Guide §10.3.1).

4.3 Weights

> **Key point:** Trying weights 1 to 3 for each model, the best is (2, 3, 1), giving R^2 0.82.

As for classification, we try each weight from 1 to 3 for each model, 27 combinations:

Weights (lr, dt, svr)	(1, 1, 1)	(1, 2, 1)	(3, 3, 1)	**(2, 3, 1)**
Mean R^2	0.809	0.818	0.818	**0.820**

The best weights give the tree, the best single model, the biggest say, and SVR, the weakest, the smallest. The gain over equal weights is small here.

4.4 One algorithm, different settings

> **Key point:** Five trees of depth 1, 3, 5, 7 and unlimited score between 0.35 and 0.74; their vote scores 0.76, above the best of them.

We can also vote over one algorithm with different hyperparameter values. Five decision trees of different depths:

max_depth	1	3	5	7	None	Vote of all five
Mean R^2	0.35	0.67	0.74	0.74	0.73	**0.76**

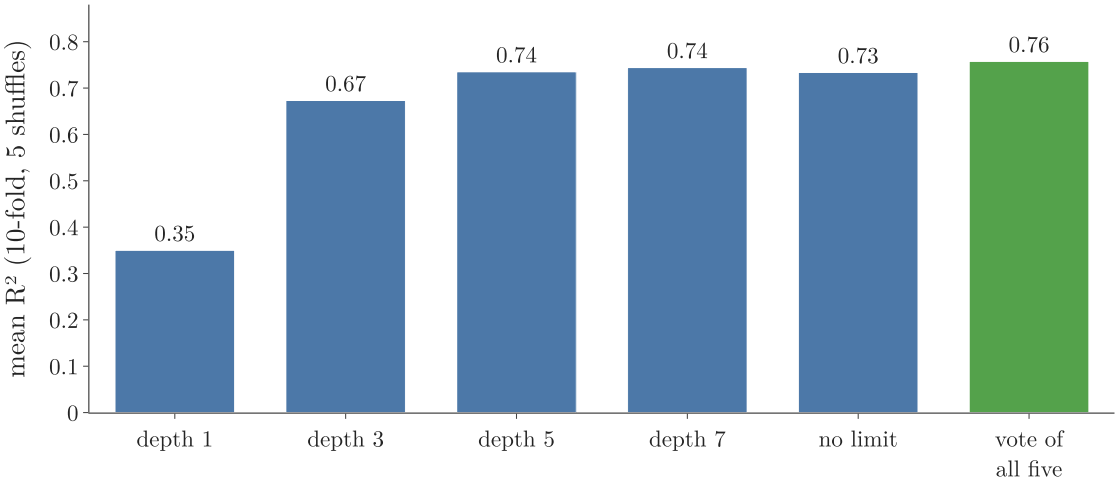


Figure 4: Mean R^2 of five decision trees of different depth on the Boston data, and of their vote

In Figure 4, the green bar is only about 0.01 above the best tree (0.757 against 0.744), a much smaller step than in Figure 3.

The vote beats every tree, even though one member (depth 1) is very weak. The gain is smaller than in section 4.2, because five trees grown on the same data make more similar mistakes than three different algorithms do.

4.5 Why we shuffle the folds

> **Key point:** Without shuffling, every test fold holds towns the model never saw, and all scores collapse.

The Boston observations are stored grouped by town: each town's observations sit together (Gilley and Pace, 1996). Plain `cv=10` cuts the folds **in order**, without shuffling, so most test

observations come from towns the model never saw in training: 453 of the 506 with folds in order, against only 19 with shuffled folds.

Figure 5 shows the difference on the rows of the data file. Each grey band in the top strip is one town. In the middle strip, the test rows of one fold cut in order form a solid block: 49 of its 51 rows come from towns that have no row in the training part. In the bottom strip, the test rows of a shuffled fold are spread over the whole file, and only 2 of 51 come from towns the model has not seen.

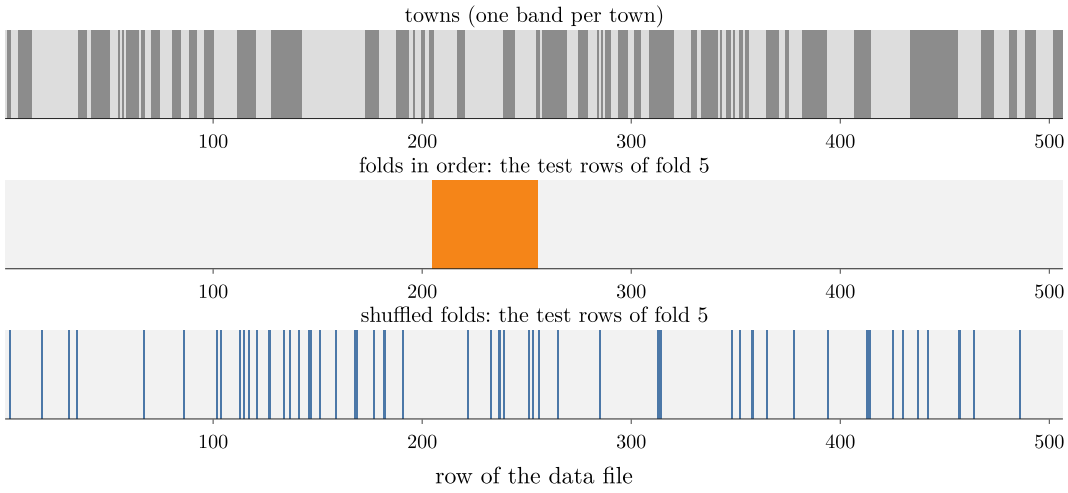


Figure 5: The 506 Boston rows in file order. Top: the towns, one band each. Middle: the test rows of one of 10 folds cut in order, a block of whole towns. Bottom: the test rows of one of 10 shuffled folds, spread across all towns

With folds in order (and the older unscaled SVR), the same models score far lower:

Model	Linear regression	Decision tree	SVR (unscaled)	Vote
R^2, folds in order	0.20	-0.07	-0.41	0.43

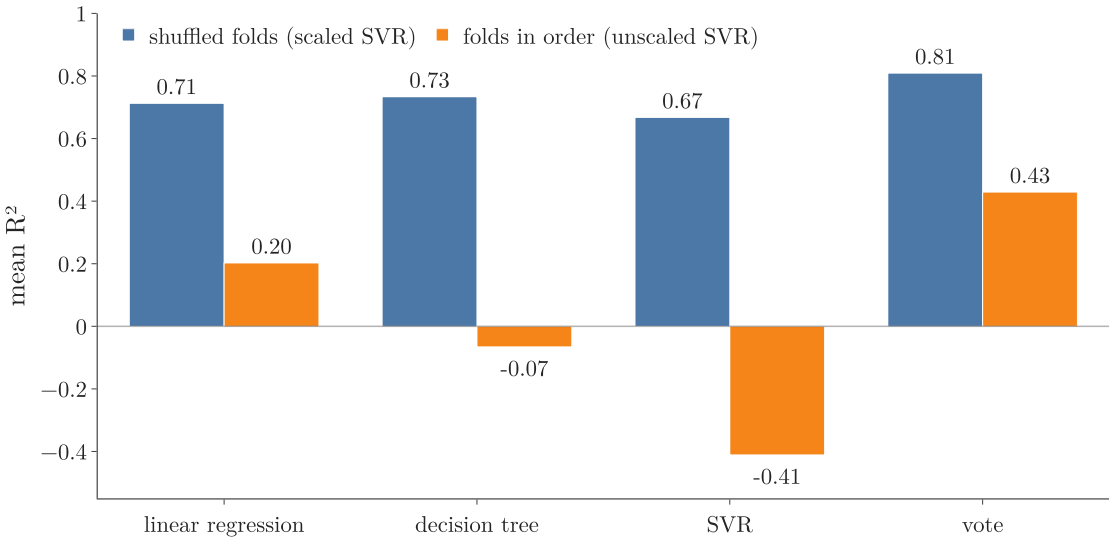


Figure 6: Mean R^2 with shuffled folds (blue) and with folds in order (orange), per model

In Figure 6, watch every orange bar fall far below its blue partner; the tree and the unscaled SVR drop below zero.

A negative R^2 means worse than always predicting the mean price.

> **Extra:** Holding out whole towns picked at random (`GroupKFold`, with the town names from the corrected data) gives scores in between: linear regression 0.55, decision tree 0.36, scaled SVR 0.61, vote 0.68. So predicting for unseen towns is genuinely harder, and the vote still beats every member there.

5. Summary

	Voting classifier	Voting regressor (G-2097)
Base models	classifiers	regressors
Aggregation	majority vote (hard) or average probability (soft)	mean of the predictions
Weights	`weights=[...]`	`weights=[...]` (weighted mean)
scikit-learn class	`VotingClassifier`	`VotingRegressor`

- A voting regressor predicts the mean (or weighted mean) of its base regressors' predictions, because a regressor outputs a number, so there is no label to vote on.
- The vote's squared error is never worse than the average member's; the more the members disagree, the bigger the gain, because the vote's error equals the members' mean squared error minus their spread around the mean (the ambiguity).
- On the noisy sine data, the voting curve smooths out the tree's jumps and scores well above the average member (0.72 against 0.58), because it follows each member only a third of the way. It does not beat SVR, so a vote is not guaranteed to beat its best member.
- On Boston, three different models (0.71, 0.73, 0.67) give a vote of 0.81 (0.82 with weights 2, 3, 1), because they are about equally good and make different mistakes that partly cancel; five trees of different depths give only 0.76, because trees grown on the same data make more similar mistakes.
- Shuffle the observations before cross-validation when the data is stored in a meaningful order, because folds cut in order test on whole towns the model never saw, and every score collapses.
- So the voting regressor is the voting classifier with a mean in place of a vote, and it pays off most when its members are about equally good and different.

6. Sources

Built from

- CampusX, "Voting Ensemble | Regression | Part 3", YouTube, https://www.youtube.com/watch?v=ut4

Other references

- Gilley, O. W. and Pace, R. K. (1996). "On the Harrison and Rubinfeld Data". *Journal of Environmental Economics and Management* 31, 403–405. Corrected data with town names: StatLib, `boston_corrected.txt` (copy in `data/`).
- Krogh, A. and Vedelsby, J. (1995). "Neural Network Ensembles, Cross Validation, and Active Learning". *Advances in Neural Information Processing Systems 7*, MIT Press, 231–238.
- scikit-learn developers. User Guide, section 10.3.1, "Parallelism". scikit-learn.org, computing/parallelism.

7. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Voting regressor (G-2097)	A regressor that predicts the mean (or weighted mean) of several trained regressors' predictions.
Ambiguity (G-2155)	How much the ensemble's members disagree: the spread of their predictions around their mean. It is the amount by which the vote's squared error beats the average member's.
n_jobs (G-1294)	scikit-learn setting for how many CPU cores to use in parallel; -1 means all.
Mean (G-1203)	The average of the values; the centre of the data.
Base model (G-260)	One of the models inside an ensemble; the ensemble combines the base models' answers into its prediction.
R^2 score (coefficient of determination) (G-1717)	1 minus the model's squared error divided by the squared error of always predicting the mean: 1 is perfect, 0 is no better than the average.
Cross-validation (G-510)	Testing a model by training and testing it several times on different parts of the training data.